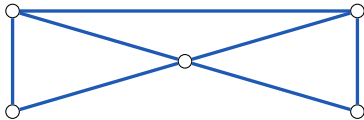


Distance in Graphs

Section 1.2

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. Distance and Metrics

How far apart are two vertices?

2. Definitions and a Few Properties

Section 1.2.1.

3. Graphs and Matrices

Section 1.2.2.

4. Graph Models and Distance

Section 1.2.3.

Metric

Let X be a set. A **metric** on X is a function

$$M: X \times X \longrightarrow \mathbb{R}_{\geq 0}$$

such that:

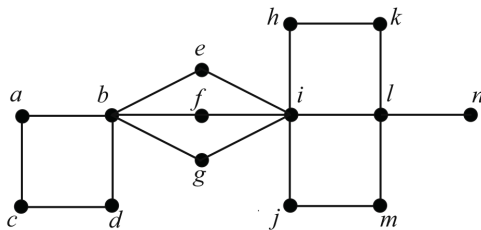
- i. $M(x, y) \geq 0$ for all $x, y \in X$, and $M(x, y) = 0$ if and only if $x = y$;
- ii. $M(x, y) = M(y, x)$ for all $x, y \in X$;
- iii. $M(x, y) \leq M(x, z) + M(z, y)$ for all $x, y, z \in X$.

Definition: Graph Distance

Graph distance

Let G be connected. The **distance** from vertex u to vertex v is the length (number of edges) of a shortest u - v path in G .

Notation: $d(u, v)$, or $d_G(u, v)$ when specifying the graph.



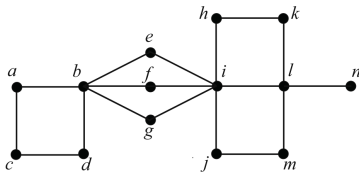
In this figure, $d(b, k) = 4$. What is $d(c, m)$?

Definition: Eccentricity

Eccentricity

Let v be a vertex of a connected graph G . The **eccentricity** of v , denoted $\text{ecc}(v)$, is defined to be the greatest distance from v to any other vertex. That is,

$$\text{ecc}(v) = \max_{x \in V(G)} \{d(v, x)\}.$$



What is $\text{ecc}(a)$?

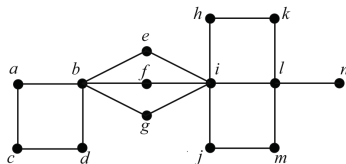
Which vertices have the greatest eccentricity? The least?

More definitions

Radius, diameter, center, and periphery

For a connected graph G :

- The **radius**, $\text{rad}(G)$, is the smallest vertex eccentricity.
- The **diameter**, $\text{diam}(G)$, is the greatest vertex eccentricity.
- The **center** is the set of vertices v with $\text{ecc}(v) = \text{rad}(G)$.
- The **periphery** is the set of vertices v with $\text{ecc}(v) = \text{diam}(G)$.



What are the radius, diameter, center, and periphery of this graph?

Question

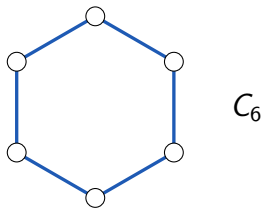
Is the diameter always bigger than the radius of a graph?

Question

Is the diameter always bigger than the radius of a graph?

Answer: No

Equality is possible.



On this graph, the diameter and radius are the same:

$$\text{diam}(C_6) = \text{rad}(C_6) = 3.$$

Theorem

Theorem 1.4

For any connected graph G ,

$$\text{rad}(G) \leq \text{diam}(G) \leq 2 \text{rad}(G).$$

Proof of Theorem

Theorem 1.4

For any connected graph G ,

$$\text{rad}(G) \leq \text{diam}(G) \leq 2 \text{rad}(G).$$

Proof. The first inequality follows directly from the definitions.

For the second, choose vertices u, v realizing the diameter: $d(u, v) = \text{diam}(G)$. Choose a central vertex c . Then

$$\begin{aligned} \text{diam}(G) &= d(u, v) \\ &\leq d(u, c) + d(c, v) \\ &\leq 2 \text{ecc}(c) = 2 \text{rad}(G). \end{aligned}$$



Remark

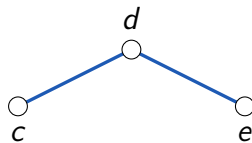
Remark

Graph distance extends to disconnected graphs as follows:

- Within a connected component, use the usual shortest-path distance.
- If two vertices lie in different components, define their distance to be ∞ .



$$d(a, b) = 1,$$



$$d(c, e) = 2,$$

$$d(a, c) = \infty.$$

Theorem

Theorem 1.5

Every graph is (isomorphic to) the center of some graph.

Proof of Theorem

Theorem 1.5

Every graph is (isomorphic to) the center of some graph.

Proof

Starting with G , form H by adding four new vertices w, x, y, z and the edges

$$\{wx, yz\} \cup \{xa : a \in V(G)\} \cup \{yb : b \in V(G)\}.$$

In H , $\text{ecc}(w) = \text{ecc}(z) = 4$ and $\text{ecc}(x) = \text{ecc}(y) = 3$.

Every $v \in V(G)$ has $\text{ecc}(v) = 2$. Thus the central vertices of H induce precisely G . □

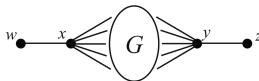


Figure 1.27. G is the center.

Theorem

Theorem 1.6

A graph G is (isomorphic to) the periphery of some graph if and only if either all its vertices have eccentricity 1, or none do.

Proof: Forward direction

Theorem 1.6

A graph G is (isomorphic to) the periphery of some graph if and only if either all its vertices have eccentricity 1, or none do.

Proof

If every vertex of G has eccentricity 1, then G is complete and is its own periphery. The one-vertex case is also immediate.

Now suppose no vertex has eccentricity 1 and $|V(G)| \geq 2$. Form H by adding a new vertex w adjacent to every vertex of G .

Each original vertex has a non-neighbor in G . In H , these vertices are at distance 2 via w , so

$$\text{ecc}_H(w) = 1, \quad \text{ecc}_H(v) = 2 \quad \text{for every } v \in V(G).$$

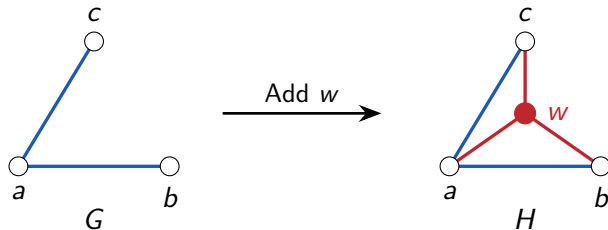
Thus the periphery of H is exactly G .



Forward Direction Diagram

Proof: the case with no eccentricity-1 vertices

Suppose $|V(G)| \geq 2$ and no vertex of G has eccentricity 1. Form H by adding a vertex w adjacent to every vertex of G . Each original vertex has a non-neighbor in G , so in H its eccentricity is 2. Since $\text{ecc}_H(w) = 1$, the periphery of H is exactly G .



$$\text{ecc}_H(w) = \text{ecc}_H(a) = 1, \quad \text{ecc}_H(b) = \text{ecc}_H(c) = 2.$$

The periphery is now $\{b, c\}$, not G .

Here $\text{ecc}_G(a) = 1$, so the hypothesis does not hold.

Proof: Reverse Direction

Theorem 1.6

A graph G is (isomorphic to) the periphery of some graph if and only if either all its vertices have eccentricity 1, or none do.

Proof

Suppose G is the periphery of a graph H . Assume, for a contradiction, that G has both a vertex of eccentricity 1 and a vertex of greater eccentricity.

Then $\text{diam}(H) \geq 2$. Choose $v \in V(G)$ with $\text{ecc}_G(v) = 1$. Since v is peripheral in H , there is a vertex w such that

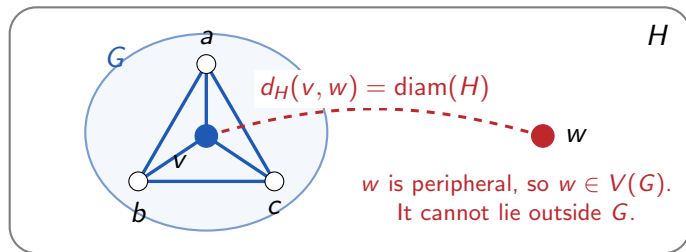
$$d_H(v, w) = \text{diam}(H) \geq 2.$$

Thus w is also peripheral in H , so $w \in V(G)$. But v is adjacent to every other vertex of G , contradicting $d_H(v, w) \geq 2$. □

Reverse Direction: Diagram

Proof

Suppose G is the periphery of H , and $v \in V(G)$ is adjacent to every other vertex of G . As in the proof, choose $w \in V(H)$ with $d_H(v, w) = \text{diam}(H) \geq 2$. Then $\text{ecc}_H(w) = \text{diam}(H)$, so w is also peripheral and must lie in G . Thus v and w are adjacent, contradicting $d_H(v, w) \geq 2$.



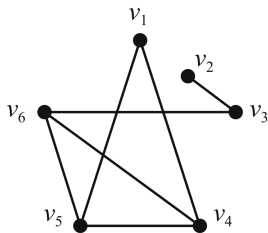
The dashed curve represents a shortest path, not an edge.

Definition: Adjacency matrix

Adjacency matrix

Let G have vertices $v_1, v_2, \dots, v_n$ in this order. Its **adjacency matrix** is the $n \times n$ matrix A with entries

$$[A]_{i,j} = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$



$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Rows and columns follow $v_1, \dots, v_6$.

Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., p. 22, Figure 1.29 and its adjacency matrix.

Remarks

- ① For simple graphs (no loops or self-edges), every entry on the main diagonal is 0.
- ② Adjacency matrices of undirected graphs are symmetric: $A = A^T$.
- ③ Different orderings of the vertices can produce different adjacency matrices.

Fact

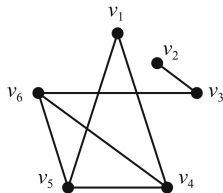
If A and B are adjacency matrices for the same graph G , there is a permutation of the vertices such that applying it to both the rows and the columns of A gives B .

Fact

If graphs G_1 and G_2 have adjacency matrices A_1 and A_2 , and applying the same permutation to the rows and columns of A_1 gives A_2 , then G_1 and G_2 are isomorphic.

Squares of Adjacency Matrices

Remark. What does A^2 tell us about our graph?



$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

$$\begin{aligned} [A^2]_{1,6} &= (0, 0, 0, 1, 1, 0) \cdot (0, 0, 1, 1, 1, 0) \\ &= (0 \cdot 0) + (0 \cdot 0) + (0 \cdot 1) + (1 \cdot 1) + (1 \cdot 1) + (0 \cdot 0) \\ &= 2. \end{aligned}$$

Notice

Each non-zero term in the above sum corresponds to a walk of length 2 from v_1 to v_6 . Hence, $[A^2]_{1,6}$ counts the number of two-edge walks from v_1 to v_6 .

Theorem

Theorem 1.7

For a graph G with ordered vertices $v_1, \dots, v_n$ and adjacency matrix A , the entry $[A^k]_{i,j}$ counts the walks of length k from v_i to v_j , for every integer $k \geq 1$.

Proof of Theorem

Theorem 1.7

For a graph G with ordered vertices $v_1, \dots, v_n$ and adjacency matrix A , the entry $[A^k]_{i,j}$ counts the walks of length k from v_i to v_j , for every integer $k \geq 1$.

Proof

Induct on k . For $k = 1$, the assertion is precisely the definition of A .

Assume the assertion holds for $k - 1$. A walk of length k from v_i to v_j has a unique penultimate vertex $v_h \in N(v_j)$. Removing its final edge leaves a walk of length $k - 1$ from v_i to v_h ; conversely, every such walk can be extended by the edge $v_h v_j$.

By the induction hypothesis, the number of length- k walks is therefore

$$\sum_{v_h \in N(v_j)} [A^{k-1}]_{i,h} = \sum_{h=1}^n [A^{k-1}]_{i,h} [A]_{h,j} = [A^k]_{i,j}.$$

This completes the induction. □

Corollary

Corollary 1.8

Let A be the adjacency matrix of G , with vertices ordered as $v_1, \dots, v_n$. If $d(v_i, v_j) = x$, then

$$[A^k]_{i,j} = 0 \quad \text{for every integer } 1 \leq k < x.$$

Question. What is the proof?

Theorem

Definition

For a graph G of order n with adjacency matrix A , and an integer $k \geq 1$, define

$$S_k = I + A + A^2 + \cdots + A^k,$$

where I is the $n \times n$ identity matrix.

Remark. All entries are nonnegative integers, so $[S_k]_{i,j} \leq [S_{k+1}]_{i,j}$ for every i, j .

Theorem 1.9

Suppose G is connected and has vertices $v_1, \dots, v_n$, with $n \geq 2$.

- ① $\text{ecc}(v_j)$ is the least positive k for which row j of S_k has no zero entries.
- ② $\text{rad}(G)$ is the least positive r for which some row of S_r has only positive entries.
- ③ $\text{diam}(G)$ is the least positive m for which every entry of S_m is positive.

Proof of Theorem

Theorem 1.9: Part 1

For a connected graph G with at least two vertices, if k is the least positive integer for which row j of S_k has no zero entries, then

$$\text{ecc}(v_j) = k.$$

Remark. Parts 2 and 3 are left as exercises.

Proof

Because every entry in row j of S_k is positive, every vertex can be reached from v_j by a walk of length at most k . Thus $\text{ecc}(v_j) \leq k$.

If $k = 1$, then $\text{ecc}(v_j) = 1$, since G has at least two vertices.

If $k > 1$, minimality of k means that row j of S_{k-1} contains a zero. The corresponding vertex is at distance greater than $k - 1$ from v_j , so $\text{ecc}(v_j) \geq k$.

Consequently, $\text{ecc}(v_j) = k$. □

Creating a New Matrix, M

Construction

Let G be a connected graph with vertices $v_1, \dots, v_n$ and adjacency matrix A . Recall that

$$S_k = I + A + A^2 + \dots + A^k.$$

Define an $n \times n$ matrix M as follows:

- 1 Set $[M]_{i,i} = 0$ for every i .
- 2 For $i \neq j$, let $[M]_{i,j}$ be the least positive integer k such that $[S_k]_{i,j} \neq 0$.

Remark

By construction, each entry records the corresponding distance:

$$[M]_{i,j} = d(v_i, v_j).$$

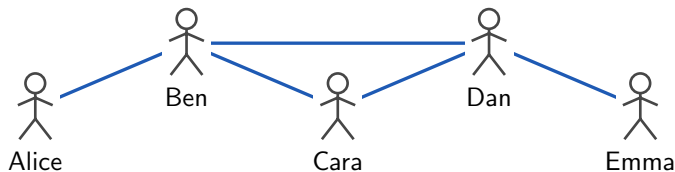
The matrix M is called the **distance matrix** of G .

Graph Model: The Acquaintance Graph

The Acquaintance Graph

Each vertex represents a person. Two vertices are joined by an edge when the corresponding people know each other.

- We may model a classroom, a neighbourhood, or the world's population.
- A path is a chain of acquaintances; distance counts the fewest links needed.
- The “six degrees of separation” claim translates to $\text{diam}(G) \leq 6$ for the worldwide graph. It is difficult to verify: the graph is huge and changes over time.



An illustrative acquaintance graph: each edge means “knows.”

Graph Model: The Hollywood Graph

The Hollywood Graph

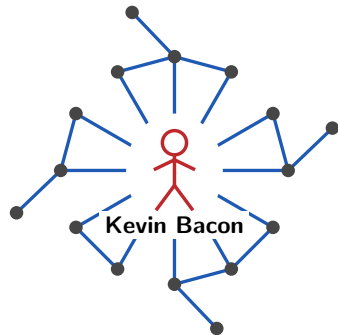
Vertices represent actors. Two actors are joined by an edge if they have appeared in a movie together.

An actor's **Bacon number** is their graph distance from Kevin Bacon: the fewest shared-movie links needed to reach him.

Bacon has number 0; his co-stars have number 1. An actor with no connection to him has number ∞ .

The **Kevin Bacon Game** challenges people to find short chains connecting actors to him—a playful example of “six degrees of separation.”

Being drawn in the middle does not by itself make Bacon a graph-theoretic center!



Schematic only; dots represent other actors.

Graph Model: Mathematical Collaboration Graph

The Mathematical Collaboration Graph

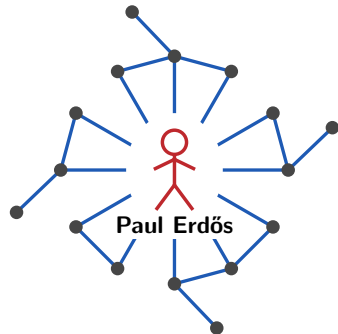
Vertices represent researchers. Two researchers are joined by an edge if they have coauthored a paper.

A researcher's **Erdős number** is their graph distance from Paul Erdős: the fewest coauthorship links needed to reach him.

Erdős has number 0, and his coauthors have number 1. Coauthors of those researchers have number 2, unless a shorter connection exists.

Erdős had hundreds of collaborators. His number system highlights connections across mathematics.

A researcher outside his connected component has Erdős number ∞ .



Schematic only; dots represent other researchers.

Small World Networks

The acquaintance graph captures the feeling that people are connected through surprisingly short chains. But the full network is enormous and constantly changing, so smaller graph models are easier to study.

To reflect this phenomenon, a small world network should have several properties:

1. Many mutual acquaintances

People should often have shared neighbours. Complete graphs have plenty of these, but they are unrealistic: not everyone knows everyone else.

2. Relatively few edges

The graph should be **sparse**: only a small fraction of all possible pairs should be adjacent. Each person knows only a small part of the population.

Small World Networks continued...

3. Short distances

The **characteristic path length** L_G is the average distance over all distinct vertex pairs. For a connected graph with $n \geq 2$ vertices,

$$L_G = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} d(v_i, v_j).$$

It is also the mean of the off-diagonal entries of the distance matrix.

4. Substantial clustering

Local groups should contain a high proportion of edges.

The book measures clustering within the closed neighbourhood $N[v]$:

$$cc(v) = \frac{|E(G[N[v]])|}{\binom{1+\deg(v)}{2}}, \quad CC(G) = \frac{1}{n} \sum_{v \in V(G)} cc(v).$$

Thus $cc(v)$ is the fraction of possible edges present in $G[N[v]]$.

Interesting Note

Erdős has a Bacon number!

If shared appearances in documentaries count, Paul Erdős has a **Bacon number of 3**, as reported in the textbook.

Erdős is the subject of the 1993 documentary *N is a Number*. Alec Guinness appears briefly in the film and has Bacon number 2, providing a link from Erdős to Kevin Bacon.

Does Bacon have an Erdős number?

The textbook reports no known chain of coauthored research papers connecting Kevin Bacon to Erdős. On that account, Bacon's Erdős number is ∞ .

A shared film appearance creates an edge in the Hollywood graph; a joint research paper creates an edge in the mathematical collaboration graph.

Trees